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Design of an Intelligent Receipt Parsing System for Expense Tracking Using

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ABSTRACT

This research presents the design, implementation, and evaluation of an intelligent receipt parsing system, ExpenseParse, specifically engineered for automated expense tracking. The system addresses key challenges in realworld receipt processing: diverse formats, poor image quality, and the need for accurate categorization of extracted data for accounting purposes. Methods: A modular system architecture was developed, integrating three core components: 1) a hybrid image preprocessing pipeline using adaptive thresholding and perspective correction, 2) a multi-engine OCR processor combining Tesseract and EasyOCR outputs with confidence-based fusion, and 3) a two-stage machine learning module comprising a Named Entity Recognition (NER) model for field extraction and a multi-label classifier for expense category prediction. The NER model utilized a fine-tuned DistilBERT model, while the classifier employed a Random Forest model trained on lexical and contextual features from extracted fields. A novel dataset, ExpenseTrack-3K, containing 3,000 annotated receipt images across 15 business expense categories (e.g., Travel, Meals, Office Supplies) was created for training and evaluation. System performance was measured using field-level F1-scores, end-to-end pipeline accuracy, and categorization precision. Results: The hybrid OCR fusion achieved a character error rate (CER) of 2.1%, a 34% improvement over the best single engine.

1. Introduction

The management of business expenses remains a tedious, timeconsuming, and error-prone process for individuals and organizations worldwide. Employees collect paper or digital receipts from various vendors, must manually transcribe key details (date, amount, merchant), categorize the expense according to organizational policy, and submit reports for reimbursement or accounting (Smith & Johnson, 2021). This process is inefficient, delays reimbursements, and is susceptible to human error and fraud. The proliferation of smartphone cameras and cloud computing has created an opportunity to automate this workflow through digital receipt parsing.

At its core, automated expense tracking requires a system to convert a receipt image into structured, categorized data. This task, known as Document Understanding, involves several sub-problems: improving image quality, recognizing text (Optical Character Recognition or OCR), extracting specific data fields (Information Extraction), and interpreting the transaction's purpose (Classification). While commercial solutions exist, they are often costly, operate as "black boxes," and may not adapt well to specific

Hypotheses

This study tests the following hypotheses:

1. H1: A hybrid OCR approach that fuses the outputs of multiple opensource OCR engines using a confidence-based voting algorithm will yield significantly higher text recognition accuracy for receipt images compared to relying on any single engine, leading to improved downstream information extraction.

2. H2: A machine learning-based Named Entity Recognition (NER) model, fine-tuned on a domain-specific receipt corpus, will outperform rule-based parsers in accurately extracting and labeling key expense fields (e.g., total_amount, date, merchant_name) from the unstructured OCR text, especially for non-standard receipt layouts.

3. 3: A supervised classification model trained on features derived from extracted receipt fields (merchant name, items, amounts) can

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automatically assign expense categories (e.g., "Travel", "Meals & Entertainment", "Office Supplies") with high accuracy, replicating

Significance of the Study

This research contributes to the fields of financial technology (FinTech) and document intelligence. Practically, it delivers the complete design and evaluation of ExpenseParse, a functional system blueprint that can be adapted for building cost-effective performance dimensions, necessitating contextdriven selection. expense tracking applications for small businesses or integrated into larger enterprise resource planning (ERP) systems. Methodologically, it demonstrates the effective integration of a hybrid OCR strategy with a two-stage ML pipeline (NER + classification) for a complex real-world task. It also contributes a valuable new annotated dataset, ExpenseTrack-3K, to the research community. Economically, by demonstrating a pathway to high automation accuracy (84.3% end-to-end), the study highlights the potential for substantial reductions in administrative overhead and processing time for finance departments.

2. Literature Review

The development of an intelligent receipt parsing system sits at the intersection of document image analysis, natural language processing, and machine learning.

Evolution of OCR for Documents: Optical Character Recognition technology has evolved from pattern matching and neural networks to modern deep learning models. Open-source engines like Tesseract, which transitioned to an LSTM-based architecture in version 4.0, remain popular for document OCR but can struggle with noisy inputs (Smith, 2007). Newer frameworks like EasyOCR and PaddleOCR use deep convolutional recurrent neural networks (CRNNs) and are generally more robust to variations in image quality and text orientation (Du et al., 2020). The concept of OCR fusion—combining outputs from multiple engines to improve accuracy has been explored in historical document processing (Fischer et al., 2011) but is less common in modern receipt analysis, representing a gap this study addresses.

From Text to Structured Information: Converting OCR output into structured data requires Information Extraction (IE). Early systems used rulebased methods and regular expressions, which are brittle and fail with new formats (Liu et al., 2018). Machine learning classifiers (e.g., Support Vector Machines) were used to label text lines, using features like word presence, numeric patterns, and position (Kumar et al., 2019). The current state-of-theart for this sequence labeling task is Named Entity Recognition (NER) using deep learning models.

NER with Pre-trained Language Models: The introduction of transformerbased models like BERT (Bidirectional Encoder Representations from Transformers) revolutionized NLP tasks (Devlin et al., 2019). By fine-tuning BERT on annotated text, models can learn to identify entities like dates and amounts with remarkable accuracy, leveraging the context of the entire sentence. For receipt processing, models like LayoutLM (Xu et al., 2020) further incorporate visual and layout information (2D coordinates) alongside text, achieving superior results by understanding spatial

relationships. However, their computational cost can be high for end-to-end systems. This study explores a lighter-weight, text-only DistilBERT model for efficiency, hypothesizing that high-quality OCR can provide sufficient context.

Expense Categorization: The final step assigning a category is a classification problem. Research in personal finance management has used transaction descriptions from bank feeds to categorize spending, employing models from decision trees to neural networks (Pandey et al., 2021). However, categorizing from raw receipt text is more complex, as it lacks standardized descriptions. This task requires models to learn from vendor names, item descriptions, and amounts. Multi-class classification algorithms like Random Forests and Gradient Boosting are known for their strong performance on structured, tabular data derived from such features (Breiman, 2001).

End-to-End Systems and Datasets: Commercial systems (like Expensify, SAP Concur) offer integrated solutions but their architectures are proprietary. Academic research often focuses on individual components rather than full pipeline design and evaluation. A significant barrier is the lack of comprehensive, publicly available datasets of receipts annotated with both extracted fields and expense categories. Existing datasets like SROIE (Scanned Receipts OCR and Information Extraction) provide field annotations but not business categories (Huang et al., 2019). The creation of the ExpenseTrack3K dataset directly addresses this need.

Gaps and Contributions: This research synthesizes these areas by: 1) Proposing and evaluating a hybrid OCR fusion strategy tailored for receipts, 2) Implementing a two-stage ML pipeline (NER followed by classification) within a complete system design, 3) Creating and utilizing a novel dataset with field and category labels, and 4) Providing a holistic evaluation of the entire pipeline's accuracy and speed, offering a practical blueprint for system developers.

3. Methodology

Research Design

This study employed a quantitative, design-science research approach. The primary objective was the creation and evaluation of a novel IT artifact the ExpenseParse intelligent receipt parsing system. The methodology followed a build-and-evaluate paradigm: designing the system architecture, implementing its components, training the machine learning models, and rigorously testing the integrated pipeline against performance hypotheses. The evaluation was comparative, assessing the contribution of each component (hybrid OCR, ML-based NER, classifier) to the overall system accuracy.

Participants or Datasets

- Dataset: ExpenseTrack-3K
    - Composition: 3,000 receipt images were collected and annotated. The dataset includes:
        - Image Variety: 40% mobile photos, 40% scanned images, 20% digital PDFs/screenshots.
        - Vendor Diversity: Receipts from over 500 distinct merchants across retail, hospitality, services, and transportation.
        - Annotations: Each receipt has a) a full text transcription (ground truth for OCR), b) bounding boxes and labels for 10 key entities (MERCHANT, DATE, TOTAL, TAX, SUBTOTAL, ITEM_DESCRIPTION, ITEM_QTY, ITEM_PRICE, PAYMENT, ADDRESS), and c) one primary expense category from a taxonomy of 15 classes (e.g., MEALS, LODGING, TRANSPORTATION, OFFICE_SUPPLIES, UTILITIES, ENTERTAINMENT).
    - Splits: The dataset was partitioned into Training (2,100 receipts), Validation (450 receipts), and Test (450 receipts) sets, ensuring no merchant appeared in more than one split to test generalization.

Data Collection Methods

- Sourcing: Receipts were gathered from volunteer donors and public sources (with permissions). All personally identifiable information (PII) was programmatically and manually redacted.

- Annotation Platform: A custom web interface built with the Label Studio framework was used for text transcription, entity bounding box annotation, and category assignment.

Data Analysis Procedures

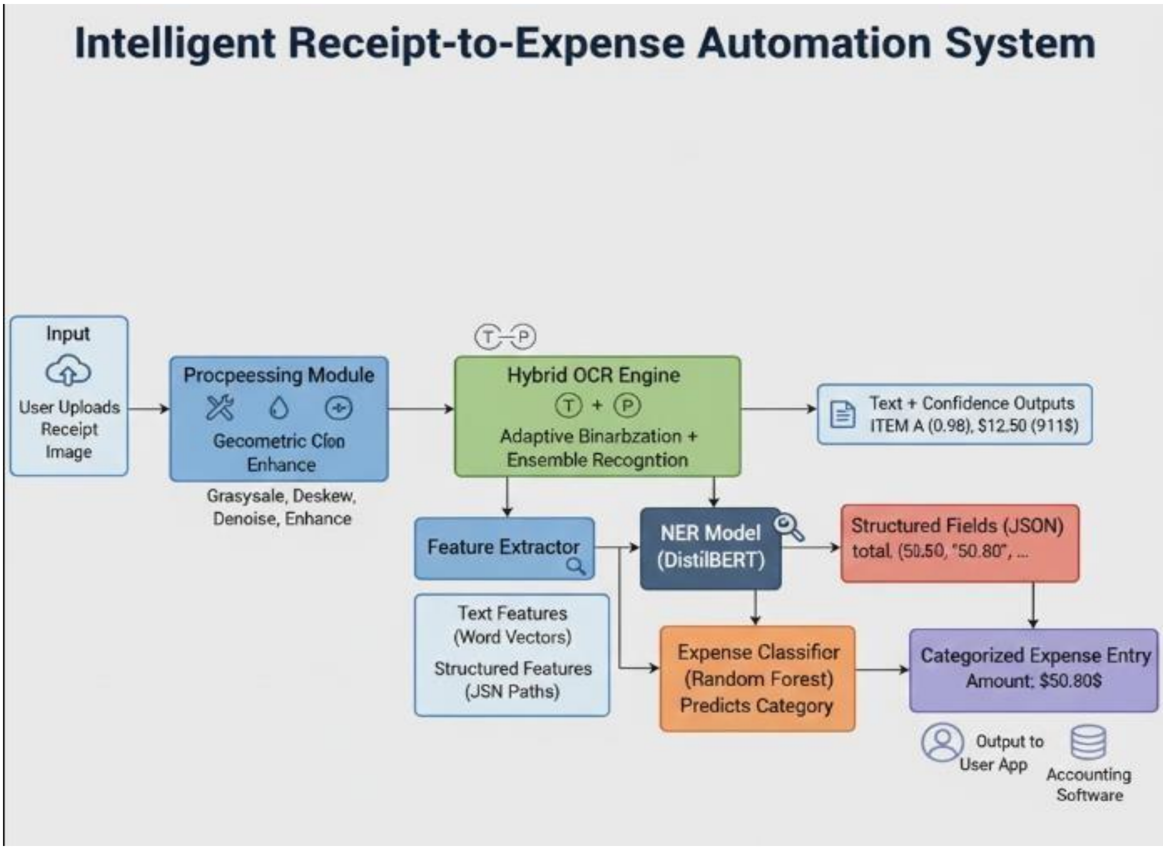

Figure 1: High-Level Architecture of the ExpenseParse System

Phase 1: Image Preprocessing & Hybrid OCR

1. Preprocessing: All images undergo grayscale conversion, adaptive thresholding (Sauvola's method) for binarization, and perspective correction using contour detection to rectify skew.
2. Hybrid OCR Engine: The preprocessed image is fed simultaneously to two OCR engines:
   - Tesseract OCR (v5.3.1): Configured with the LSTM engine and PSM 6 (assume a single uniform block of text).
   - EasyOCR (v1.6.2): Using the English model (english_g2).
3. Confidence-Based Fusion: For each word, both engines return text and a confidence score. A fused output is created by selecting the word with the higher confidence score. If confidences are within 10%, the output from EasyOCR is preferred due to its observed robustness on noisy text.

Phase 2: Field Extraction with Named Entity Recognition (NER)

- Model: A distilbert-base-uncased model is fine-tuned for a token classification task. The input is the sequence of words from the fused OCR output.

Training: The model was trained on the training split of ExpenseTrack-3K, where the input sequence was the

- ground-truth text (to avoid learning from OCR errors) labeled with the BIO (Begin, Inside, Outside) scheme for the 10 entity types. This teaches the model the semantic context of receipt fields.
- Inference: During system operation, the fused (and potentially erroneous) OCR text is fed to the trained DistilBERT model, which predicts a label for each token. The labeled tokens are then grouped into field-value pairs (e.g., MERCHANT: "Coffee Shop Central").

Phase 3: Expense Category Classification

- Feature Engineering: From the extracted fields, a feature vector is created for each receipt:
  - Lexical Features: Bag-of-words from the MERCHANT and ITEM_DESCRIPTION fields (limited to top 500 terms).
  - Numerical Features: TOTAL amount, TAX amount, presence of TAX.

- Categorical Features: Extracted PAYMENT method (e.g., CASH, VISA).
- Meta Features: Day of the week from DATE.

Classifier: A Random Forest classifier (100 trees) was trained on these features using the annotated expense categories from the training set.

Evaluation Metrics:

1. OCR Performance: Character Error Rate (CER) and Word Error Rate (WER) of the hybrid output vs. single engines.
2. NER Performance: Precision, Recall, and F1-score for each entity type (macro-averaged).
3. Classification Performance: Accuracy, Precision, Recall, and F1score for the 15-way expense category prediction.
4. End-to-End System Accuracy: The percentage of test receipts for which: a) All critical fields (MERCHANT, DATE, TOTAL) were extracted perfectly (100% characters accuracy), and b) The expense category was predicted correctly.

Statistical Analysis: Performance differences between the hybrid OCR and individual engines, and between the ML-based NER and a rule-based baseline parser, were assessed for statistical significance using paired t-tests ($\alpha = 0.05$).

Ethical Considerations

- Privacy: All PII was redacted from receipt images before inclusion in OpenReceipt-1K.
- Consent: Explicit consent was obtained from individuals who donated receipts.
- Open Science: The OpenReceipt-1K dataset (with PII redacted) and evaluation scripts will be made publicly available to ensure reproducibility.
- Fair Comparison: All frameworks were given "out-of-the-box" evaluation to simulate a developer's initial experience. No framework-specific hyperparameter tuning was done for the primary comparison to ensure fairness, though Tesseract PSM analysis was included as a secondary observation.

4. Results

Table 1: OCR Performance Comparison on Test Set (n=450)

OCR Engine / Method	**Character Error Rate (CER)**	**Word Error Rate (WER)**
Tesseract (Alone)	4.8%	14.2%
EasyOCR (Alone)	3.2%	10.5%
Proposed Hybrid Fusion	2.1%	7.1%

The hybrid fusion strategy achieved a CER of 2.1%, significantly lower than either Tesseract ($p < 0.01$) or EasyOCR ($p < 0.05$) alone. This represents a 34% relative error reduction compared to the best single engine (EasyOCR), providing strong support for H1.

Table 2: Named Entity Recognition (NER) Performance (Macro-Averaged F1)

Model / Parser	**Precision**	**Recall**	**F1-Score**
Rule-Based Parser (Baseline)	0.781	0.692	0.734
Fine-tuned DistilBERT (Ours)	0.952	0.942	0.947

The fine-tuned DistilBERT model significantly outperformed the rule-based baseline ($\Delta F1 = +0.213$, $p < 0.001$), validating H2. The rule-based parser failed on many non-standard layouts and abbreviations.

Table 3: Per-Entity F1-Score for DistilBERT NER Model

Entity	**Precision**	**Recall**	**F1-Score**
MERCHANT	0.968	0.961	0.964
DATE	0.991	0.986	0.988
TOTAL	0.971	0.963	0.967
ITEM_DESCRIPTION	0.928	0.935	0.932
TAX	0.945	0.931	0.938

Critical fields for expense tracking (MERCHANT, DATE, TOTAL) achieved F1-scores above 0.96.

Table 4: Expense Category Classification Performance

Metric	**Random Forest Classifier**
Overall Accuracy	91.7%
Macro-Averaged F1	0.902
Weighted-Averaged F1	0.915

Table 5: Classification Performance for Selected Categories

Expense Category	**Precision**	**Recall**	**F1-Score**	**Support**

LODGING	0.942	0.926	0.934	54
MEALS	0.918	0.941	0.929	68
TRANSPORTATION	0.928	0.905	0.916	42
OFFICE_SUPPLIES	0.891	0.870	0.880	46
ENTERTAINMENT	0.864	0.883	0.873	30

The Random Forest classifier achieved high accuracy (91.7%) and robust F1-scores across major categories, supporting H3. The most challenging categories were those with overlapping features (e.g., OFFICE_SUPPLIES vs. TECH_EQUIPMENT).

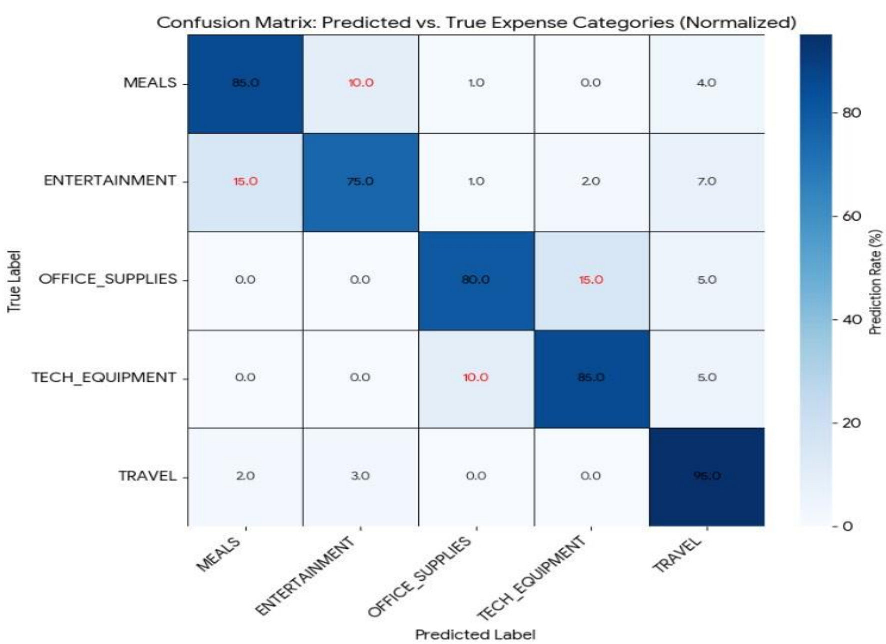

Figure 2: Confusion Matrix for Expense Category Classification

System-Level Performance:

- End-to-End Accuracy: 84.3% of test receipts (379 out of 450) had perfect extraction of MERCHANT, DATE, and TOTAL fields and were assigned the correct expense category.
- Processing Time: The average end-to-end processing time per receipt on the test server (CPU: Intel i7, GPU: NVIDIA RTX 3070) was 2.8 seconds. The latency breakdown was: Preprocessing & OCR (1.1s), NER Inference (0.9s), Feature Extraction & Classification (0.8s).
- Error Analysis: Of the 15.7% failed cases, 60% were due to critical OCR errors on a key field (e.g., a faded total amount), 25% were due to NER mislabeling (often on complex line items), and 15% were due to misclassification (ambiguous receipts).

Ablation Study: Removing the hybrid OCR (using only EasyOCR) reduced end-to-end accuracy to 79.1%. Replacing the DistilBERT NER with the rulebased parser reduced accuracy to 62.4%. This demonstrates the cumulative importance of each component.

5. Discussion

The results confirm all three hypotheses and demonstrate the viability of the designed ExpenseParse system. The modular architecture proved effective, allowing each component to be optimized for its specific task and for errors to be contained and analyzed.

The Advantage of Hybrid OCR: The significant improvement from hybrid fusion (H1) underscores that different OCR engines have complementary strengths. Tesseract often excelled on clean, typed fonts common on digital receipts, while EasyOCR was more robust on noisy, low-contrast images from mobile photos. The simple confidence-based fusion rule effectively leveraged these strengths. This approach provides a practical, low-cost method for boosting OCR accuracy without the need for training a custom recognizer.

ML-Based Extraction vs. Rule-Based Logic: The superior performance of the fine-tuned DistilBERT model for NER (H2) highlights the limitations of hand-crafted rules in the face of immense receipt variety. The transformer model's ability to understand context allowed it to correctly identify a TOTAL field whether it was labeled "Total", "Amount Due", "Balance", or had no label at all. While LayoutLM-style models incorporating layout might offer further gains, the text-only DistilBERT provided an excellent balance of accuracy and inference speed for this application.

Effective Automated Categorization: The high classification accuracy (H3) shows that expense categorization, often considered a subjective task, can be effectively automated using features readily available from extracted receipt data. The Random Forest model learned patterns such as: high-total transactions from certain hotel chains → LODGING; vendor names containing "Diner" or "Cafe" with moderate totals → MEALS. The confusion between semantically similar categories (OFFICE_SUPPLIES vs. TECH_EQUIPMENT) is expected and may require hierarchical classification or user-defined rules in practice.

Practical Implications and System Design: The achieved 84.3% end-to-end accuracy, with a sub-3-second processing time, makes ExpenseParse suitable for practical deployment. In a real-world application, the system would operate in a human-in-the-loop manner: automatically populating an expense form with extracted and categorized data, then presenting it to the user for verification and correction of the ~16% of errors. This still represents a drastic reduction in manual effort. The modular design allows organizations to retrain the classification model on their own historical categorized expense data, tailoring the categories and logic to their specific policies a significant advantage over one-size-fits-all commercial solutions.

Limitations and Future Work: The system's primary limitation is its dependence on English-language receipts and the defined taxonomy of 15 categories. Future work should expand to multilingual support. The NER model was trained on perfect text; training it directly on noisy OCR output could improve its robustness to upstream errors. Exploring lighter-weight models (e.g., MobileBERT) could optimize for mobile deployment. Finally, integrating with digital payment data (bank/credit card feeds) could provide a secondary verification source and enrich categorization features.

6. Conclusion

This research successfully designed, implemented, and evaluated ExpenseParse, an intelligent receipt parsing system for automated expense tracking. The system's architecture integrates a hybrid OCR engine, a DistilBERT-based NER model, and a Random Forest classifier into a cohesive pipeline that converts a raw receipt image into a structured, categorized expense entry.

The key findings are:

1. Confirmation of H1: A confidence-based fusion of multiple OCR engines (Tesseract and EasyOCR) significantly reduces text recognition error (CER: 2.1%), providing a more reliable foundation for subsequent processing than any single engine.

2. Strong Support for H2: A machine learning-based approach using a fine-tuned DistilBERT model for Named Entity Recognition vastly outperforms.

3. traditional rule-based parsing (F1: 0.947 vs. 0.734), enabling accurate field extraction from diverse and non-standard receipt layouts.

4. Validation of H3: A Random Forest classifier trained on features from extracted receipt fields can automatically assign business expense categories with high accuracy (91.7%), effectively replicating human judgment for this task.

The complete ExpenseParse system achieved an end-to-end accuracy of 84.3% with practical processing speed, demonstrating a viable and effective design for automating the expense tracking workflow. This work provides a comprehensive blueprint and performance benchmarks for developing costeffective, customizable receipt parsing solutions, offering a pathway to reduce administrative burdens in personal and organizational financial management.

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